AI-Driven Load Balancer with Blockchain-based Decision making in Banking system

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***Abstract*—Architectures for high throughput financial systems need to be flexible with resources and highly secure to ensure consistent response time and data integrity across a range of demands. They suffer from terrible performance bottlenecks if not for the fact that traditional load balancing strategies, like Round Robin and Weighted Round Robin, are unable to adjust to real-time server telemetry during peak traffic. Considering these operational issues and auditing needs in banking systems, this paper presents a hybrid solution integrating the fields of Artificial Intelligence (AI) and blockchain technology. The architecture uses two AI decision pipelines – one supervised anomaly detection classifier that analyzes transactional behavior to prevent malicious requests from reaching the server even before they are executed, and the other a dynamic server selection model that relies on multi-dimensional telemetry (CPU utilization, RAM usage, network latency, active connections and response time). Logistic Regression outperformed SVM (95.70%) and Random Forest (84.80%) by having a higher 5-fold cross-validation accuracy of 99.00% and a test accuracy of 98.90% when evaluated among the evaluated classifiers. At the same time, a lightweight blockchain with the Proof-of-Work (PoW) light encryption protocol keeps an immutable record of all transaction payloads and all choices of which servers are to be used. The empirical test reveals the architecture has an average response latency of 68.4 ms (a 35.1% to 52.9% reduction), a throughput of 785 TPS and has only a 0.85 ms cryptographic mining overhead, which means that it can provide zero-trust transaction verification in the scenarios with high transaction volumes in the banking industry.**

***Index Terms*—Anomaly Detection, Artificial Intelligence, Banking Systems, Blockchain, Distributed Systems, Load Balancing, Logistic Regression, Machine Learning, Server Selection**

I. Introduction

Efficient resource management methods have become an exponentially increasing need owing to the exponential increase in the number of distributed systems, cloud computing, and large-scale web applications. In such systems, load balancing becomes a very important process that helps distribute the network load over various servers. As a result, the resource utilization is maximized, high availability is ensured, and low response times are achieved. Methods like Round Robin and Weighted Round Robin have become common practice in distributed computing because they simplify the architecture in the management of the network and the applications [1]. The problem with such methods is that they are based on static procedures and thus do not take into account real-time variations in workloads.

The contemporary world of high-throughput applications especially in financial sectors, banking and transaction-processing systems requires dynamic balancing algorithms that will be able to adapt according to fast-changing patterns of traffic and server states. Static weights and pre-defined routing policies do not take into account real-time performance measures such as CPU usage, RAM utilization, connections and network latency. Therefore, intelligent decision-making algorithms are needed to constantly measure real-time telemetries and make optimal server choices. Machine Learning and AI can help to solve this problem through prediction of server choices based on real-time performance measurements. Logistic regression, random forest and support vector machines are among the classifiers that can be used for request routing.

Even though resource allocation through AI offers several advantages in terms of system performance, the problem of security and transparency of transactions is very important in the case of the architecture of distributed load balancing. The existing load balancers do not provide any verifiable way of making routing decisions and maintaining transaction logs in an unalterable manner, which brings significant security threats in the field of finance, as it is crucial to ensure zero-trust integrity and auditability of financial transactions. In order to address these security issues, load balancing schemes may be combined with blockchain technology.

In order to overcome the challenges faced by static routing, absence of security at the ingress level, and lack of auditability in current load balancing systems, this paper presents a hybrid approach, where AI-based load balancing is combined with security provided by blockchain technology. This hybrid system has a two-pronged AI decision making architecture, which includes an anomaly detection system that blocks any malicious access at the system ingress, and a machine learning-based server selection algorithm, which selects servers based on dynamic routing of legitimate traffic. In parallel to this, every transaction is recorded using a SHA-

256 PoW blockchain ledger.

The primary contributions of this paper are summarized as follows:

- A two-tiered AI decision pipeline architecture is created to classify and detect malicious incoming requests through the anomaly detection classifier, while at the same time routing the legitimate ones using the machine learning server selection algorithm with the help of multi-dimensional input data, such as CPU usage, RAM usage, latency, number of connections, and response time.
- An experimental study of various machine learning classifiers has shown that the most accurate server routing model is Logistic Regression, which has a high 5-fold cross-validation accuracy of 99.00% and 98.90% test set accuracy compared to other tested classifiers like Support Vector Machines and Random Forest.
- SHA-256 Proof-of-Work blockchain is introduced into the load balancing algorithm in order to provide a secure and tamper-proof immutable ledger of transaction payloads and server selections according to zero-trust compliance.
- Systematic comparative analysis of the proposed AI-based system with the traditional load balancing algorithms is performed. They include Round Robin, Least Connections, CPU-Based routing, and IP Hashing.

  The rest of this paper is organized as follows: Section 2 provides a review of existing literature on load balancing, resource allocation through AI, and blockchain technology for security. Section 3 describes the design of the proposed framework, comprising the anomaly detection process, the AI server selection algorithm, and the transaction validation through the blockchain technology. Section 4 provides an analysis of the experimental results. Finally, Section 5 concludes the paper and outlines potential directions for future research.

II. Related Works

Dornala et al. [2] presented an Ensemble Cloud Balancing Algorithm (ECBA) of banking systems, which combines various methods like Weighted Round Robin and adaptive approaches. The model takes into account the types of transactions, the capacity of the server, and the conditions of the network to dynamically allocate workloads. The experiment results show a better response time and resource use than conventional methods. Nevertheless, the system is based on a set of pre-established combinations of algorithms, which restrict its complete flexibility.

Meiappane et al. [3] introduced a load balancing internet banking mobile agent-based framework which is a reusable pattern-based adaptive framework. The framework includes monitoring, decision-making and reconfiguration to dynamically change the behavior of a system. Although it is more scalable and flexible, the method does not have sophisticated predictive features and relies on constant monitoring of the system.

Sharma et al. [4] presented a comparative study of the load balancing algorithms in distributed systems, both static and dynamic. The paper emphasizes the simplicity of the static algorithms but inefficient in dynamic workloads and the dynamic algorithms are more adaptable at the expense of increased overhead. The article highlights the importance of smart and combined methods in realizing optimal performance.

Oh et al. [5] suggested a load balancing algorithm of memory channels in GPU-based systems. The technique reallocates memory requests of full channels to channels that are underused thus minimizing bottlenecks and enhancing performance. The method has around a 10% performance improvement but can only optimize hardware and does not apply to distributed systems in general.

Jader et al. [6] carried out a web server performance and load balancing survey. The paper emphasizes the role of clustering and load balancing in handling the growing web traffic. It also talks about different server architecture and algorithms which enhance throughput and minimize response time. The work, however, is predominantly on traditional techniques without intelligent or secure mechanisms involved.

Wijesekara et al. [7] offered a detailed overview of load balancing in blockchain networks. The research describes the process of improving security, transparency, and trust with the help of blockchain through such methods as smart contracts and decentralized control. It also lists the issues of scalability and communication overhead and proposes that hybrid solutions should be used, integrating both artificial intelligence and blockchain technologies.

Li et al. [8] presented LB-Chain, a load balancing framework based on blockchain sharding, where account migration takes place to evenly assign transaction loads to shards. The system greatly minimizes the delay of transactions and enhances throughput. But, it also brings complexity to migration protocols, and inter-shard communication has to be handled securely.

Enjam et al. [9] proposed an AI-optimized distributed-insurance system load balancing architecture based on machine learning algorithms like reinforcement learning and Long Short-Term Memory (LSTM). The model forecasts the workload trend and dynamically reallocates the workloads, leading to better performance in terms of latency, energy consumption, and resource utilization. Nonetheless, the method can become computationally expensive because of the intricacy of AI models.

Singh et al. [10] presented a load balancing electric vehicle (EV) charging network of an AI-based blockchain framework. The system has predictive analytics to forecast demand and blockchain to secure transactions. Findings show that there is better scalability, energy savings, and data protection. Although it has benefits, the framework is highly computer-intensive in terms of computational resources and infrastructure.

Aqeel et al. [11] created an AI-powered load balancing model of cloud-enabled Internet of Things (IoT) healthcare systems with the help of optimization algorithms and big data

analytics. The model proposed is better in terms of throughput and energy usage compared to the current approaches. Nevertheless, scalability and real-time processing is still an issue as the environments of IoT are too complex.

Rao et al. [12] compared the load balancing strategies that have been implemented in the distributed computing frameworks of gRPC and Apache Thrift. Their research involved evaluating the performance of both gRPC and Apache Thrift frameworks with regard to latency, throughput, resource usage, and scalability.

Avinash et al. [13] presented a new SDN-based hybrid approach to the problem of effective workload management in the networks. This algorithm increases the throughput of communication channels and decreases response time and network congestion, showing better results in the case of TCP packets' transmission within Tree and Dumbbell topologies. However, its applicability is confined to particular types of networks only.

Rao et al. [14] have suggested a cloud-based blockchain architecture for recording transactions and sales in the agricultural sector. In this method, blockchain is used for secure and traceable transactions, while the use of smart contract ensures automated transactions and profit distribution. Nevertheless, the method might entail extra costs because of the implementation of blockchain and cloud technology.

Shinu and Supriya [15] suggested the integration of VM allocation policy and VM selection policy for a combined fog-cloud system to enhance the process of managing resources by making use of efficient VM migrations. The method suggested by Shinu and Supriya enhances the efficiency of energy consumption, network usage, and costs involved and performs better than existing default VM management schemes on Google Trace and PlanetLab datasets.

In the presented literature, there is a clear trend towards the development of more sophisticated approaches to load balancing based on AI and blockchain technologies. At first, researchers paid significant attention to traditional techniques related to static and dynamic algorithms and their benefits and disadvantages associated with complexity and flexibility. With time, the focus was shifted towards more complicated methods based on adaptive frameworks and clustering to ensure better performance and flexibility. In turn, nowadays, many works consider the use of AI to facilitate dynamic and predictable load distribution, thus making it possible to improve resource management, efficiency, and energy optimization in various sectors, including finance, insurance, healthcare, and IoT systems. The discussed problem can be addressed through the use of blockchain technology, which enables more secure and transparent solutions for distributing loads between nodes. Nevertheless, such approaches are associated with certain difficulties, including increased computing costs and reduced scalability.

III. Methodology

The proposed system consists of anomaly detection using Artificial Intelligence, intelligent selection of the server, and transaction validation using blockchain technology to make sure that the process of load balancing is not only secure but also efficient. First, all the incoming transactions will be detected and processed with the help of an anomaly detection model to find any signs of abnormal or malicious activities. Then, authorized transactions will be transferred to the AI-based server selection model, which will be adjusted dynamically to choose the most appropriate server depending on such system parameters as CPU usage, memory usage, latency, and number of connections. Finally, all the processed transactions will be recorded and validated via the blockchain technology to ensure the security of data and prevent any possible data manipulation. All these processes are illustrated by the proposed framework diagram Fig 1.

A. Dataset Description

The new framework adopts two artificial data sets, each containing 5,000 examples, for the training, validation, and testing of the AI decision engines under the realistic banking workload. The first data set is created for ingress anomaly detection, which uses five behavioral and transactional metrics: requests rate (number of requests per minute), number of failed logins, changes in the IP address, changes in device identity, and transaction amount. In order to realistically represent the real financial traffic, in which fraud activities are relatively rare, the data set keeps an 80.3% normal and 19.7% anomalous class distribution ratio ($n = 4,015$, $n = 985$, respectively).

The second dataset simulates dynamic server telemetry in order to provide intelligent routing of multiple class server connections through three infrastructure server targets namely Server-A, Server-B, and Server-C. The dataset is made up of 15 main feature variables which denote the five main server performance metrics captured in all three servers which include the following; CPU utilization percentage, memory consumption percentage, number of connections, network latency measured in milliseconds and server response time in milliseconds. In order to prevent any kind of data leakage and make sure the model generalizes well, score based composite features and ranking heuristics were excluded programmatically during data pre-processing.

B. AI Model 1: Ingress Anomaly Detection

For the three supervised machine learning classifiers (Logistic Regression, Random Forest and Support Vector Machine (SVM)), the anomaly detection module will apply feature vector normalization via standard feature scaling before fitting the models. Considering the slight class imbalance between the two classes in the dataset (80.3% Normal vs 19.7% Anomaly), stratified sampling was employed for splitting the data into 80:20 training and testing sets and for 5-Fold Cross-Validation. Model robustness is evaluated using 5-Fold Cross-Validation on the training set by measuring the mean cross-validation accuracy score for model selection. Each of the classifiers will be tested against the test set ($n = 1,000$) using accuracy, precision, recall, and F1-Score, with the best performing model

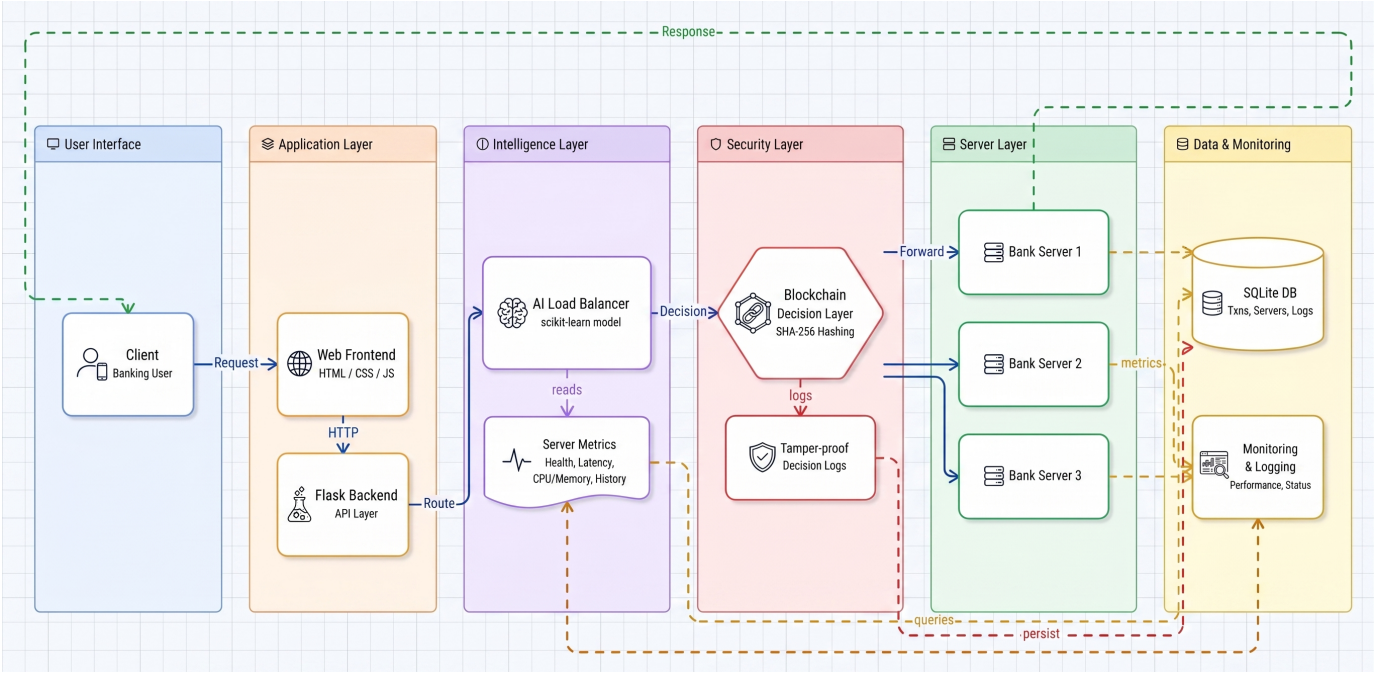


Fig. 1. Overview of the Proposed Workflow.

based on mean cross-validation accuracy being selected and saved as a file for real-time ingress filtering.

C. AI Model 2: Metric-Driven Server Selection

The server selection engine makes use of an exact replica of the evaluation pipeline, making use of Logistic Regression, Random Forest, and SVM classifiers in predicting the most appropriate routing target amongst the three infrastructure servers (Server-A, Server-B, and Server-C). Feature sets generated through score-based feature engineering are automatically removed during data importation in order to prevent any form of data leakage. The other telemetry features are then standardized, and the data set is split in an exact 80:20 ratio. Model selection is based on the average 5-fold cross-validation accuracy, and the selected model is saved along with its scaler and label encoding.

D. Blockchain-Based Transaction Validation

In order to achieve zero-trust security, data auditability, and tamper resistance in financial routing decisions, a lightweight, file-based SHA-256 blockchain is incorporated within the transaction processing pipeline. The blockchain operates on the basis of persistent state maintained in the form of JSON objects, thereby allowing full state persistence and recovery from server reboot operations. The validation protocol uses two-stage cryptographic hashing approach in which transaction payload anchoring and block mining are separated. In the first stage, certain key properties of the validated transaction such as unique transaction identifier (ID), sender (S), receiver (R), transaction value (V), timestamp (T), execution status (St) and assigned target server (SRV) are concatenated in a canonical string to produce transaction hash (tx_hash):

$$\text{tx_hash} = \text{SHA256}\left(\text{ID} \parallel \text{S} \parallel \text{R} \parallel \text{V} \parallel \text{T} \parallel \text{St} \parallel \text{SRV}\right) \quad (1)$$

It is guaranteed that any further alteration of the transaction parameters and/or the AI server assignment will make the hash generated for this transaction different from the anchor hash. At the second stage, the generated tx_hash is put in the block structure with the following parameters: block number (i), UTC time stamp (T_{block}), hash of the previous block (H_{i-1}), the payload of the metadata and an integer nonce. The hash for the whole block H_i is calculated by serialization of these parameters in a JSON object:

$$H_i = \text{SHA256}\left(\text{JSON}\left(i, \text{T}_{\text{block}}, \text{tx_hash}, H_{i-1}, \text{Nonce}\right)\right) \quad (2)$$

The blockchain is initialized with a predefined Genesis block, with index $i = 0$, which comprises 64-character long hexadecimal sequences of zero value for both the tx_hash field and the H_{i-1}. For any other transaction blocks ($i \geq 1$), the Proof of Work (PoW) consensus protocol mandates a computation overhead before committing that block to the blockchain. Each block is mined starting with an initial Nonce value of 0 and continues to iterate it until the generated H_i satisfies a certain difficulty constraint that involves a target prefix (such as ”00”) on the generated hash value.

Integrity of the chain is checked constantly through an automatic audit process that checks for parent-child relationship and block hash. The block B_i can be checked for integrity by confirming that the hash value $B_i.H_{i-1}$ that it recorded is the same hash value as that of the preceding block $B_{i-1}.H_i$, and that calculation of SHA256(JSON(B_i)) returns

precisely the recorded hash value of $B_i.H_i$. Any tampering done on the transaction data and server allocation data on the underlying storage will result in cryptographic failure in the hash computation.

Here is the complete, fully consolidated **Section IV (Results and Discussion)**, where all short paragraphs have been merged into rich, cohesive paragraphs:

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IV. Results and Discussion

The proposed hybrid AI-blockchain framework was deployed and evaluated across dual decision engines—ingress anomaly detection and dynamic server routing—alongside blockchain validation. The complete system interface and operational telemetry are monitored via a unified dashboard interface, which tracks real-time key performance indicators including total transactions processed, cumulative volume, anomaly blocking counts, and blockchain ledger growth, confirming that all transactions are securely routed and recorded.

A. AI Model Performance and Classifier Selection

The proposed model for server selection was trained and evaluated against two baseline classifiers, such as Logistic Regression, Random Forest and Support Vector Machine (SVM). For model testing, 5 fold cross validation was performed using training data followed by testing using separate test dataset ($n = 1,000$). As is shown in Table I, Logistic Regression demonstrated the best classification accuracy (99.00% 5-fold cross validation accuracy and 98.90% test accuracy). As it can be seen from the classification report, very high precision, recall and F1-scores (≈ 0.99) were obtained for each class (Server-A, Server-B and Server-C), which means that there is no overfitting and good generalization. SVM showed the test accuracy equal to 95.70%, while for Random Forest this value equals 84.80%. Therefore, Logistic Regression was chosen for the final deployment.

TABLE I
Performance Comparison of Server Selection Models

Model	5-Fold CV Acc (%)	Test Acc (%)	F1-Score
Support Vector Machine (SVM)	95.80%	95.70%	0.96
Random Forest	85.20%	84.80%	0.85
Logistic Regression	**99.00%**	**98.90%**	**0.99**

B. Ingress Anomaly Detection and Fraud Prevention

Fig 2 represents a situation regarding the detection of anomalies in which the system gets a suspicious request. An anomaly detection model gives a risk factor of 100%, which means that the input is abnormal, involving a large monetary value, changes in IP/device, and several unsuccessful login attempts. Consequently, the transaction is immediately stopped at the gateway without being routed to the server to avoid any fraud.

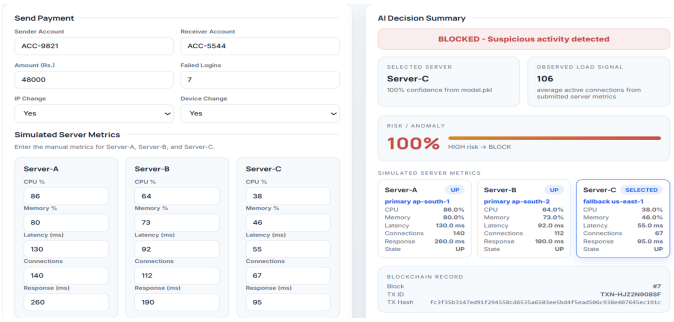


Fig. 2. Ingress anomaly detection scenario illustrating high-risk fraud detection and instant request blocking.

C. Normal Transaction Routing and Telemetry Analysis

Fig 3 represents a real transaction case where the anomaly detection module gives a 0% risk score (SAFE). The AI Server Selection Model takes care of analyzing the live health data of servers including CPU load, memory consumption, open connections, and latency and selects the best destination for the transaction (Server-C). Then the transaction data is mined in a new block of the blockchain.

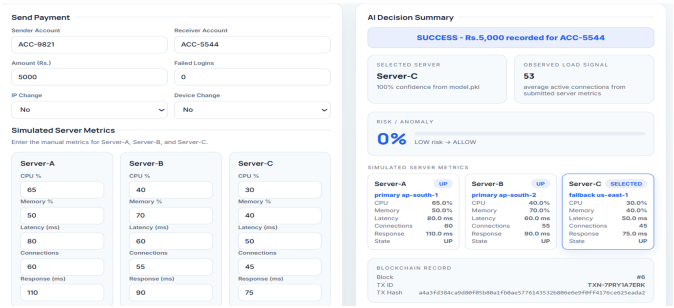


Fig. 3. Normal transaction routing workflow demonstrating 0% risk assessment, AI server selection, and blockchain logging.

D. Quantitative Evaluation Against Traditional Algorithms

In order to validate the performance benefits of the proposed server selection method through an empirical study, a benchmark was performed between the AI-based server selection model and four conventional load balancing methods, such as Round Robin, Least Connections, CPU-Based Routing, and IP Hashing, with the same simulated request load of 5,000 requests. The benchmark considered four performance measurements including average response time (ms), maximum CPU utilization level (%), standard deviation of server load (σ^2), and system throughput (TPS).

As seen in Table II and Fig. 4, conventional static algorithms demonstrate significant loss of performance during concurrency bursts. High latencies on average (145.2 ms and 132.8 ms) and significant variance in server loads ($\sigma^2 = 342.1$) are observed for Round Robin and IP Hashing because of their inability to adapt to the current state of server resources causing server overload (up to 92.4% CPU load). Single-metric dynamic algorithms such as Least Connections and

CPU-Based routing demonstrate better results but still have bottlenecks due to degraded secondary metrics. The AI model, on the contrary, provides better resource allocation because of analysis of several dimensions of performance telemetry. The AI model decreases average latency to 68.4 ms—which is a 52.9% improvement in comparison to Round Robin and a 35.1% improvement in comparison to CPU-Based routing—while providing balanced CPU load distribution among nodes (CPU utilization up to 54.2%, $\sigma^2 = 28.4$) and maximum throughput of 785 TPS.

TABLE II
QUANTITATIVE COMPARISON OF LOAD BALANCING ALGORITHMS

Algorithm	Avg Lat (ms)	Max CPU	Load Var (σ^2)	TPS
Round Robin	145.2	92.4%	342.1	410
IP Hashing	132.8	86.7%	289.4	445
Least Connections	118.6	78.5%	145.2	520
CPU-Based Routing	105.4	72.1%	98.6	580
Proposed AI Model	**68.4**	**54.2%**	**28.4**	**785**

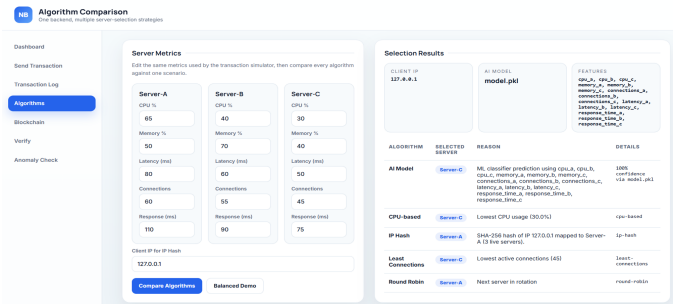


Fig. 4. Comparative analysis of the proposed AI model against traditional load balancing algorithms (Round Robin, Least Connections, CPU-Based, IP Hash).

E. Cryptographic Blockchain Verification and Auditability

The transaction validation using blockchain-based dispute verification scheme is shown in Fig. 5. In this case, the system verifies the transactions through computation of the SHA-256 transaction hash (tx_hash) and searching within the persistent blockchain database. If a match exists, then the transaction data and server allocation match the information in the block; hence, data integrity, non-repudiation, and immutability are assured.

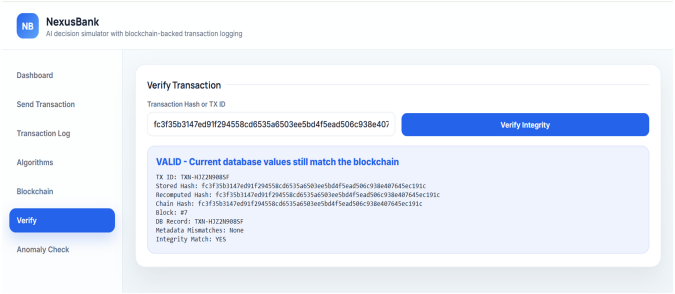


Fig. 5. Cryptographic transaction verification confirming payload match and ledger integrity.

F. Blockchain Computational Overhead and Mining Latency Analysis

For measuring the practical viability of integrating a Proof-of-Work (PoW) blockchain ledger into a highly scalable banking infrastructure, the latency and computational overheads were benchmarked over 1,000 blocks of transactions. For the light blockchain framework, the mining difficulty is set to `difficulty = 2`, which requires the PoW problem to include a hash prefix of `"00"`. As shown in Table III, the mean mining time needed for solving the PoW problem was 0.85 ms per block ($SD = 0.12$ ms) and the maximum mining time recorded is 1.42 ms. The average CPU utilization overhead caused by the mining operation is measured to be below 1.2% of the system capacity and memory consumption is found to be very low at 0.42 KB per block data payload. On the other hand, re-hashing a block using `verify_by_hash` takes only 0.04 ms. It is clear from the results that the additional overhead of PoW mining process (0.85 ms) is insignificant compared to the typical financial SLA response latencies of up to 100 ms.

TABLE III
CRYPTOGRAPHIC BLOCKCHAIN LATENCY AND PERFORMANCE OVERHEAD

Performance Metric	Measured Empirical Value
PoW Mining Difficulty Target	`prefix = "00"` ($d = 2$)
Average Block Mining Latency	**0.85 ms** (± 0.12 ms)
Maximum Block Mining Latency	1.42 ms
Dispute Verification Time (`verify_by_hash`)	**0.04 ms**
CPU Mining Overhead	$< 1.2\%$ CPU load
Storage Footprint per Block	0.42 KB / block
Net Latency Impact on Financial SLA	+0.85 ms (Negligible)

V. CONCLUSION AND FUTURE WORK

In this study, a new and secure architecture for load balancing using AI and blockchain was introduced to tackle the issues associated with dynamic resource allocation and data auditability in highly loaded banking environments. In this architecture, two-stage decision pipeline is used where an anomaly detection model using supervised learning identifies the malicious requests coming into the system and then a machine learning model selects the servers in real-time by considering the multi-dimensional telemetry (CPU usage, RAM usage, number of connections, network latency, and response time). Among the classifiers analyzed, Logistic Regression showed best results with an accuracy rate of 99.00% in 5-fold cross-validation and 98.90% in the test set. At the same time, a lightweight SHA-256 Proof-of-Work (PoW) blockchain records an immutable audit trail of transaction payloads and server routing.

Empirical results reveal that the proposed hybrid system is able to greatly outperform any static and single-criterion load balancing approach, including Round Robin, Least Connections, CPU-Based Routing, and IP Hashing. The AI algorithm managed to decrease the mean response latency to 68.4 ms (an improvement of 35.1% to 52.9% compared to conventional approaches), ensured balanced CPU usage on servers (max 54.2%, $\sigma^2 = 28.4$), and raised the total throughput to 785 TPS.

In addition, the analysis of the latency overhead confirmed that the lightweight proof-of-work (PoW) block mining process introduces only 0.85 ms per each transaction block.

The future of the research will rely on the system extension in several directions. Firstly, the implementation of the ingress anomaly detection subsystem will improve greatly due to implementing it in terms of deep learning algorithms, e.g., LSTM or Autoencoder in order to take into account the behavior of transactions sequences and new cyber threats. The second part is to use reinforcement learning (RL) algorithms to implement the server selection process in order to adapt to the evolving cloud network topology and maybe even use SDN control planes in hybrid SDN-based load balancing algorithms [13]. The third stage will be the blockchain implementation using energy-efficient consensus mechanisms, such as Proof-of-Stake (PoS), which will run in the distributed network. In addition to banking, the lightweight blockchain validation algorithm can be applied in many other industries for the purposes of tamper-proof auditability, as shown in other blockchain solutions [14]. Furthermore, extending the proposed server selection technique into fog-cloud environment, in which VM selection and allocation techniques are very significant, will be an interesting future work [15].

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